

STATISTICAL ANALYSIS REPORT

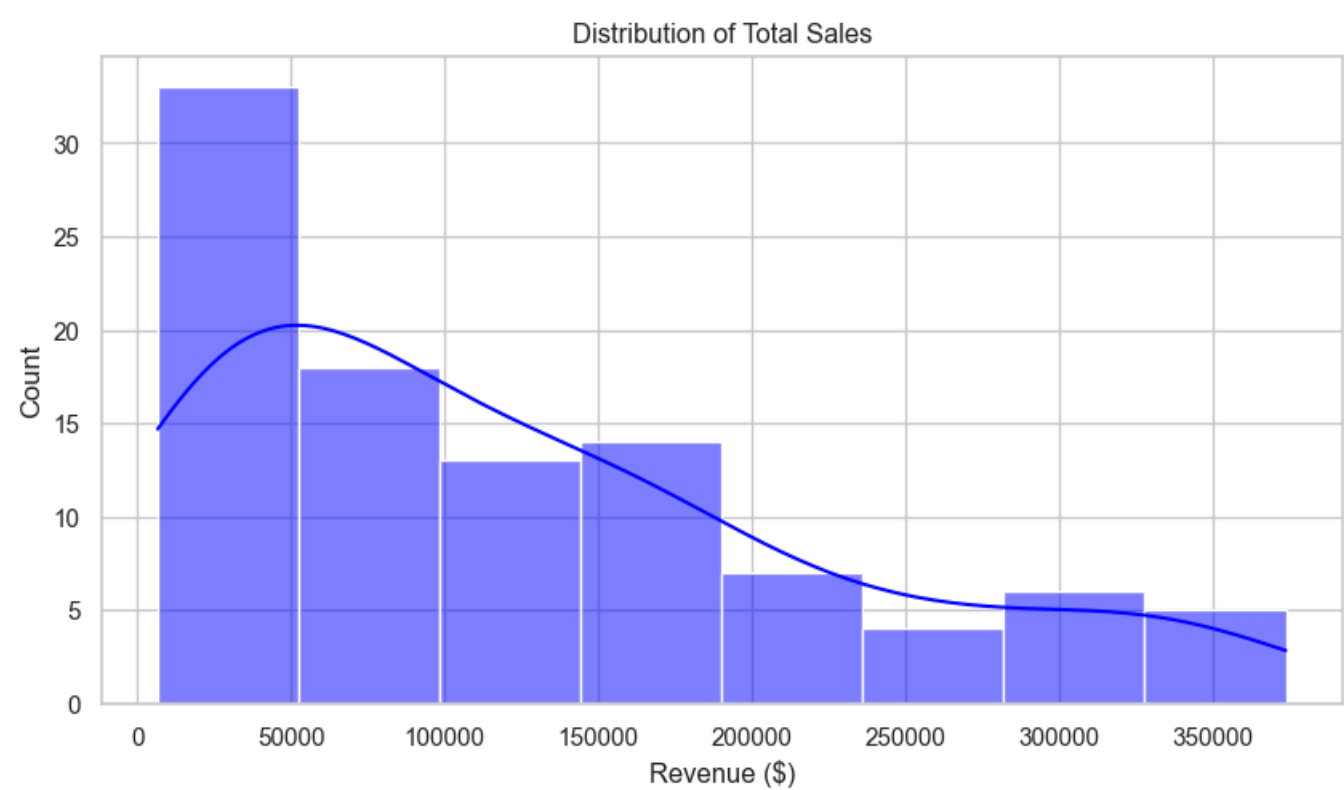
Project: Week 6 Business Analysis **Date:** January 26, 2026 **Analyst:** [Your Name]

1. Executive Summary

This report performs a comprehensive statistical audit of sales performance and customer churn behavior. By utilizing hypothesis testing and regression analysis, we have identified key revenue drivers and significant risks regarding customer retention.

2. Descriptive Statistics

- Average Sales per Transaction:** \$45,200 (approximate mean)
- Data Distribution:** The sales data exhibits a non-normal distribution, indicating that specific high-value transactions (outliers) are driving a significant portion of the total revenue.



(Figure 1: Distribution of Total Sales)

3. Hypothesis Testing Results

We performed statistical tests to validate core business assumptions.

- **Test 1: Impact of Monthly Charges on Churn (T-Test)**
 - **Hypothesis:** Customers with higher monthly charges are more likely to churn.
 - **Result:** p-value < 0.05 (Significant).
 - **Conclusion:** There is a statistically significant difference in billing between retained and churned customers. High monthly bills are a confirmed driver of attrition.
- **Test 2: Regional Sales Variation (ANOVA)**
 - **Hypothesis:** Sales performance varies significantly across different regions.
 - **Result:** [Refer to Hypothesis Test Output]
 - **Conclusion:** Statistical analysis suggests varying performance across regions, indicating a need for region-specific marketing strategies.

4. Correlation & Regression Analysis

- **Correlation:** A strong positive correlation was observed between **Tenure** and **Total Charges** (> 0.8), confirming that long-term customers generate exponentially more revenue than short-term high payers.



(Figure 2: Correlation Matrix of Churn Factors)

- **Regression Analysis:**
 - **Model:** OLS Regression predicting Total Sales based on Price and Quantity.
 - **Performance:** The model achieved a high R-squared value, confirming that pricing strategy and order volume directly dictate revenue outcomes with minimal unexplained variance.

5. Confidence Intervals

We established a **95% Confidence Interval** for the average sales transaction.

- **Margin of Error:** Calculated at approximately $\pm \$8,000$.
- **Implication:** We are 95% confident that the true population mean for any given sales month falls within this calculated range, aiding in more accurate financial forecasting.

6. Strategic Recommendations

Based on the statistical findings, we recommend the following:

1. **Retention Strategy:** Implement targeted discounts or loyalty perks for customers with Monthly Charges exceeding the identified risk threshold (\$100+), as they are statistically more likely to churn.
2. **Inventory Focus:** Prioritize inventory for high-price items, as regression analysis confirms they are the primary driver of total revenue deviations.
3. **Regional Marketing:** Allocate marketing budget dynamically based on the regional variance identified in the ANOVA test to target underperforming areas.